

Review Exercise

2. Factorize the following expressions:

(i) $4x^3 + 18x^2 - 12x$

$$\begin{aligned} &4x^3 + 18x^2 - 12x \\ &= 2x(2x^2 + 9x - 6) \end{aligned}$$

(ii) $x^3 + 64y^3$

$$\begin{aligned} &x^3 + 64y^3 \\ &= (x)^3 + (4y)^3 \\ &= (x + 4y)[(x)^2 - (x)(4y) + (4y)^2] \\ &= (x + 4y)(x^2 - 4xy + 16y^2) \end{aligned}$$

(iii) $x^3y^3 - 8$

$$\begin{aligned} &x^3y^3 - 8 \\ &= (xy)^3 - (2)^3 \\ &= (xy - 2)[(xy)^2 + (xy)(2) + (2)^2] \\ &= (xy - 2)(x^2y^2 + 2xy + 4) \end{aligned}$$

(iv) $-x^2 - 23x - 60$

$$\begin{aligned} &-x^2 - 23x - 60 \\ &= -[x^2 + 23x + 60] \\ &= -[x^2 + 20x + 3x + 60] \\ &= -[x(x + 20) + 3(x + 20)] \\ &= -(x + 20)(x + 3) \end{aligned}$$

(v) $2x^2 + 7x + 3$

$$\begin{aligned} &2x^2 + 7x + 3 \\ &= 2x^2 + 6x + x + 3 \\ &= 2x(x + 3) + 1(x + 3) \\ &= (x + 3)(2x + 1) \end{aligned}$$

(vi) $x^4 + 64$

$$\begin{aligned} &x^4 + 64 \\ &= (x^2)^2 + (8)^2 + 2(x^2)(8) - 16x^2 \\ &= (x^2 + 8)^2 - (4x)^2 \\ &= (x^2 + 8 + 4x)(x^2 + 8 - 4x) \\ &= (x^2 + 4x + 8)(x^2 - 4x + 8) \end{aligned}$$

(vii) $x^4 + 2x^2 + 9$

$$\begin{aligned} &x^4 + 2x^2 + 9 \\ &= x^4 + 9 + 2x^2 \\ &= (x^2)^2 + (3)^2 + 2(x^2)(3) - 6x^2 + 2x^2 \\ &= (x^2 + 3)^2 - 4x^2 \\ &= (x^2 + 3)^2 - (2x)^2 \\ &= (x^2 + 3 + 2x)(x^2 + 3 - 2x) \\ &= (x^2 + 2x + 3)(x^2 - 2x + 3) \end{aligned}$$

(viii) $(x + 3)(x + 4)(x + 5)(x + 6) - 360$

$$\begin{aligned} &(x + 3)(x + 4)(x + 5)(x + 6) - 360 \\ &\quad 3 + 6 = 9 = 4 + 5 \\ &= [(x + 3)(x + 6)][(x + 4)(x + 5)] - 360 \end{aligned}$$

$$\begin{aligned} &= [x(x + 6) + 3(x + 6)][x(x + 5) + 4(x + 5)] - 360 \\ &= [x^2 + 6x + 3x + 18][x^2 + 5x + 4x + 20] - 360 \\ &= [x^2 + 9x + 18][x^2 + 9x + 20] - 360 \end{aligned}$$

Let $y = x^2 + 9x$

$$\begin{aligned} &= (y + 18)(y + 20) - 360 \\ &= y(y + 20) + 18(y + 20) - 360 \\ &= y^2 + 20y + 18y + 360 - 360 \\ &= y^2 + 38y \\ &= y(y + 38) \\ &= (x^2 + 9x)(x^2 + 9x + 3) \\ &= x(x + 9)(x^2 + 9x + 3) \end{aligned}$$

(ix) $(x^2 + 6x + 3)(x^2 + 6x - 9) + 36$

$$(x^2 + 6x + 3)(x^2 + 6x - 9) + 36$$

Let $y = x^2 + 6x$

$$\begin{aligned} &= (y + 3)(y - 9) + 36 \\ &= y(y - 9) + 3(y - 9) + 36 \\ &= y^2 - 9y + 3y - 27 + 36 \\ &= y^2 - 6y + 9 \\ &= y^2 - 3y - 3y + 9 \\ &= y(y - 3) - 3(y - 3) \\ &= (y - 3)(y - 3) \\ &= (y - 3)^2 \\ &= (x^2 + 6x - 3)^2 \end{aligned}$$

2. Find HCF and LCM:

(i) $4x^3 + 3x^2, 8x^2 + 16x$

$$4x^3 + 3x^2 = 4x^2(x + 3) = \boxed{2 \cdot 2} \cdot \boxed{x} \cdot x \cdot (x + 3)$$

$$8x^2 + 16x = 8x(x + 2) = 2 \cdot \boxed{2 \cdot 2} \cdot \boxed{x} \cdot (x + 2)$$

$$HCF = 2 \cdot 2 \cdot x$$

$$HCF = 4x$$

$$LCM = [2 \cdot 2 \cdot x][2 \cdot x \cdot (x + 3) \cdot (x + 2)]$$

$$LCM = 8x^2(x + 3)(x + 2)$$

(ii) $x^3 + 3x^2 - 4x, x^2 - 4x + 3$

Factorization

$$\begin{aligned} &x^3 + 3x^2 - 4x \\ &= x[x^2 + 3x - 4] \\ &= x[x^2 + 4x - x - 4] \\ &= x[x(x + 4) - 1(x + 4)] \\ &= x[(x + 4)(x - 1)] \\ &= x(x + 4)(x - 1) \end{aligned}$$

$$\begin{aligned} &x^2 - 4x + 3 \\ &= x^2 - 3x - x + 3 \\ &= x(x - 3) - 1(x - 3) \\ &= (x - 3)(x - 1) \end{aligned}$$

Now

$$x^3 + 3x^2 - 4x = x(x + 4)\boxed{(x - 1)}$$

$$x^2 - 4x + 3 = (x - 3)\boxed{(x - 1)}$$

$$HCF = x - 1$$

$$LCM = [(x - 1)][x \cdot (x + 4) \cdot (x - 3)]$$

$$LCM = x(x - 1)(x + 4)(x - 3)$$

$$(iii) x^2 + 8x + 16, x^2 - 16$$

Factorization

$$\begin{aligned} & x^2 + 8x + 16 \\ &= x^2 + 8x + 16 \\ &= x^2 + 4x + 4x + 16 \\ &= x(x + 4) + 4(x + 4) \\ &= (x + 4)(x + 4) \end{aligned}$$

$$\begin{aligned} & x^2 - 16 \\ &= (x)^2 - (4)^2 \\ &= (x + 4)(x - 4) \end{aligned}$$

Now

$$x^2 + 8x + 16 = (x + 4)(x + 4)$$

$$x^2 - 16 = (x + 4)(x - 4)$$

$$HCF = x + 4$$

$$LCM = [(x + 4)][(x + 4) \cdot (x - 4)]$$

$$LCM = (x + 4)^2(x - 4)$$

$$(iv) x^3 - 9x, x^2 - x - 6$$

Factorization

$$\begin{aligned} & x^3 - 9x \\ &= x[x^2 - 9] \\ &= x[(x)^2 - (3)^2] \\ &= x[(x + 3)(x - 3)] \\ &= x(x + 3)(x - 3) \end{aligned}$$

$$\begin{aligned} & x^2 - x - 6 \\ &= x^2 - 3x + 2x - 6 \\ &= x(x - 3) + 2(x - 3) \\ &= (x - 3)(x + 2) \end{aligned}$$

Now

$$x^3 - 9x = x(x + 3)(x - 3)$$

$$x^2 - x - 6 = (x - 3)(x + 2)$$

$$HCF = x - 3$$

$$LCM = [(x - 3)][x \cdot (x + 3) \cdot (x + 2)]$$

$$LCM = x(x - 3)(x + 3)(x + 2)$$

$$LCM = x[(x)^2 - (3)^2](x + 2)$$

$$LCM = x(x^2 - 9)(x + 2)$$

4. Find square root by factorization and division method of the expression $16x^4 + 8x^2 + 1$

Factorization Method:

$$\begin{aligned} 16x^4 + 8x^2 + 1 &= (4x^2)^2 + 2(4x^2)(1) + (1)^2 \\ 16x^4 + 8x^2 + 1 &= (4x^2 + 1)^2 \end{aligned}$$

Taking square root on both sides

$$\begin{aligned} \sqrt{16x^4 + 8x^2 + 1} &= \sqrt{(4x^2 + 1)^2} \\ \sqrt{16x^4 + 8x^2 + 1} &= \pm(4x^2 + 1) \end{aligned}$$

Division Method:

$$\begin{array}{r} 4x^2 + 1 \\ 4x^2 \overline{) 16x^4 + 8x^2 + 1} \\ \underline{\pm 16x^4} \\ 8x^2 + 1 \\ 8x^2 \overline{) 8x^2 + 1} \\ \underline{\pm 8x^2 \pm 1} \\ 0 \end{array}$$

$$\text{Square Root} = \pm(4x^2 + 1)$$

5. Huria is analyzing the total cost of her loan, modeled by the expression $C(x) = x^2 - 8x + 15$, where x represents the number of years. What is the optimal repayment period for Huria's loan?

$$C(x) = x^2 - 8x + 15$$

$$C(x) = x^2 - 5x - 3x + 15$$

$$C(x) = x(x - 5) - 3(x - 5)$$

$$C(x) = (x - 5)(x - 3)$$

Put $C(x) = 0$ for optimal repayment period:

$$C(x) = 0$$

$$(x - 5)(x - 3) = 0$$

Either

$$x - 5 = 0$$

or

$$x - 3 = 0$$

$x = 5$,
Repayment period for Huria's loan will be 3 years or 5 years.

$x = 3$

Muhammad Tayyab (GHS Christian Daska)